RTEU COMPUTER ENGINEERING DEPARTMENT
CEN322-Internet of Things Midterm Exam

This report contains **2 parts**. Each part is worth **50 points**, totaling **100 points**. Good luck.

Question No	Part1	Part2	TOTAL
Points	50	50	100

OVERALL EVALUATION

Part No	Topic	LO	Max Points	Earned Points
1	IoT System Architecture and Hardware–Software Components	LO.1, LO.2	50	
2	ESP32-S3 Based System Design and Wireless Sensor Integration	LO.1, LO.2	50	
TOTAL			**100**	

Project Assignment Format

Cover Page

- Project Title
- Student's Name and Surname
- Student Number
- Date

Table of Contents

1. Introduction
2. Project Description and Objectives
3. Methods and Materials Used
4. Arduino Codes and Explanations
5. Results and Discussion
6. Conclusion
7. Appendices (if any)
8. References

◆ PART 1 – SYSTEM EXPLANATION (LO1 – 50%)

1. Introduction

- What is the purpose of the project?
- What problem does it solve?
- Why did you choose this project?

2. Project Description and Objectives

- What is the system?
- What are its main components?
- What are the project goals?

3. System Architecture

3.1 Overall IoT Architecture

- Device layer (ESP32-S3, sensors)
- Network layer (Wi-Fi, MQTT, HTTP, etc.)
- Application layer (server, app, dashboard)

Include a **block diagram** showing how data flows.

3.2 Hardware Explanation

- ESP32-S3 role
- Sensors used
- Other components
- Why were they selected?

3.3 Software Explanation

- Development environment (Arduino IDE or ESP-IDF)
- Libraries used

- General program structure

3.4 Network and Communication

- Which protocol is used? (Wi-Fi, MQTT, HTTP, etc.)
- How does data move from sensor to server?

◆ PART 2 – DESIGN & IMPLEMENTATION (LO2 – 50%)

4. Materials and Tools Used

4.1 Hardware

Example:

- ESP32-S3
- Sensor(s)
- Breadboard
- Jumper wires
- Resistors
- Microphone / Speaker (if used)

4.2 Software

- Arduino IDE / ESP-IDF
- Required libraries

5. Circuit Design

- Include circuit diagram
- Explain connections
- Show how sensor connects to ESP32-S3

6. Arduino Codes and Explanation

6.1 Program Structure

Explain:

- setup()
- loop()
- Main logic

6.2 Code Snippet Example

```
int sensorValue = analogRead(A0);
// Reads analog sensor value
```

Explain:

- What the code does
- Why it is needed

Include wireless communication code if used.

7. Results and Discussion

- Does the system work?
- Is sensor data correct?
- Does Wi-Fi communication work?
- What problems did you face?
- How did you solve them?

8. Conclusion

- Did you achieve your objectives?
- What can be improved in the future?

9. Appendices

Include:

- .ino code files
- Circuit diagram file
- README file explaining how to run the project

10. References

List books, articles, or websites used.